Assessing journal and institution influence through reputation recirculation via citations

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Abstract

Traditional citation-based metrics, like the Journal Impact Factor, measure popularity but are vulnerable to manipulation and fail to account for the prestige of the citing source. Eigenvector-based methods (also called spectral ranking) address the desirability of accounting for the influence of citing journals although they remain within the reflexive journal-to-journal citation system and can be manipulated by orchestrating citations.

We introduce a novel method, the Journal-Institution Reputation Circulation (JIRC) algorithm, that recirculates reputation between journals and institutions via a two-mode citation network to derive more robust influence metrics. We construct two normalized citation matrices: one from institutions to journals (Matrix A) and the other from journals to institutions (Matrix B). The algorithm solves for the dominant eigenvector of the combined system $v = P_B^T P_A^T v$, where v is the institutional influence vector, and derives the journal influence vector w as $w = P_A^T v$. We apply JIRC to a large dataset from the ESI Field of Economic & Business.

The resulting journal and institution classifications show strong construct validity, aligning with established reputations. Moreover, when used to screen a sample of 330 researchers—equally dividing them into a "Control" group (Nobel laureates and prestigious award winners) and a "Test" group (authors with highly cited papers but no major awards)—JIRC effectively disentangles influence from notoriety. The top tier of researchers was predominantly from the Control group, while the bottom tier was overwhelmingly composed of members from the Test group. The JIRC algorithm thus provides a robust, manipulation-resistant tool for assessing influence at the journal, institutional, and individual researcher level, offering a significant advancement over purely citation-count-based methods.

Keywords: Scientometrics, Research Evaluation, Citation Analysis, Reputation, Eigenvector Centrality, Pinski-Narin, PageRank, Journal Influence, Institutional Ranking

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1 Introduction

The review, evaluation and assessment of a researcher's work is a high-stakes activity that shapes professional success in publication, recognition, grant funding, appointment, advancement and financial rewards. The temptation to boost, curate, misrepresent, orchestrate or fake research is accordingly high, countered by regulations, moral/ethical values, and the opprobrium attached to violation of norms (Merton, 1996). The interplay of temptation and morality leads to the publication of works designed primarily to inflate performance indicators, rather than report findings that advance research. The proportion of such 'aberrations' is not well characterised, but the informed estimate by Heathers (2025) that 1/7 of the literature is 'fake' corresponds currently to around 700,000 works per year. Efficient and accurate methods to screen the research literature for aberrations at this scale would support well-focused, systematic, organised scepticism that is so important to scientific progress (Ziman, 2001, Chapt. 9).

Some aberrations in scientific publications are revealed by material within the document, including fake papers (Else and Van Noorden, 2021), plagiarism (Long et al, 2009), data fabrication (Brown et al, 2018), falsification of methods (Simonsohn, 2013) and image manipulation (Bik et al, 2016). For others, potential aberrations are revealed by bibliometric and reference metadata, including quesaguilloComparingUniversityRankings2010tionable author-editor relations (Banks, 2009), guest authorship (Khezr and Mohan, 2022), coerced references (Straub and Anderson, 2009), hyper-production (Ioannidis, 2024), citation cartels (Safón et al, 2025), and orchestration (Evdaimon et al, 2024).

Bibliometric studies are not normally designed to detect aberrations but rather to identify and rate worthy research or reveal the practice of science. Nevertheless, some are applicable to large scale screening for aberrations if one moves from an appreciative to investigatory perspective: e.g. scrutinising high publication or citation counts rather than offering accolades. As discussed by Tahamtan and Bornmann (2022) in their proposal for the Social Systems Citation Theory (SSCT), a bibliometric approach does not need to deal with humans and their motives but rather may begin with and focus on the communication network of research. Once identified, aberrant works may be set aside (e.g. in systematic reviews, Munn et al, 2021), perused through the lens of forensic bibliometrics (McIntosh and Vitale, 2024), or subject to deeper but time-consuming textual and peer review (Kilicoglu, 2018). Aberrant work may also help select subjects for ethnographic or psychosocial studies of researchers' individual motives and collective behaviours (Martinson et al, 2005).

Effective large-scale screening for anomalies using journal-level bibliometric indicators has been demonstrated by Kojaku et al (2021). Their CIDRE (Citation Donors and Recipients) algorithm detects anomalous citation patterns between journals, revealing candidates for possible exclusion from journal indexing services. Chakraborty et al (2020) address a similar problem using sets of features to identify anomalies in computer science journals. Evdaimon et al (2024) exhibit author-level screening using the referencing information of tens of millions of articles to quantify the extreme values of field-specific indicator distribution functions that can isolate specific patterns of referencing orchestration. Ioannidis (2024), Ioannidis et al (2024), and Ioannidis and Maniadis (2023) identify many indicators that potentially herald large-scale manipulation. Ibrahim et al (2025) exhibit an author's citation concentration index, c^2, the largest number n of papers that cite the author at least n times, showing that it can be used as a warning about manipulation and also revealing how pre-print servers may be exploited.

These and similar studies demonstrate that the tails of some frequency distributions (FDs) derived from populations of aggregated citations or references include a higher proportion of aberrant publications. An alternative to examining extrema of FDs is to screen a reference or citation before aggregation, for example by applying a weight. An article's field-weighted citation index (FWCI) and relative citation ratio (RCR) are familiar indicators that weight citations to adjust for benign referencing variations between fields of research. The reputable citation (RC) method of Safón et al (2025) uses an exogenous rating of citing author's institutions to scale citations prior to aggregation at the journal or author level. Applied to the field of Mathematics, a large gap between aggregations of scaled and unscaled citations delineates authors whose elevated citation profile may have little to do with the modest impact and quality of their work.

While the exogenous institution ranking used in the RC method helps to insulate the indicator from anomalous publications, it is challenging to find the large-scale, independent, field-specific institution rankings that the method uses. Accordingly in this paper we use spectral ranking over a multi-partite citation network to establish indicators of journal and institution prestige. These are then conferred on citations by works hosted by the respective sources and institutions. The raw and scaled citation counts are then used to compute an author esteem indicator. In this way we exhibit a screen that promises to reveal low-esteem authors who contribute anomalous works to the research literature.

The paper is organised as follows. First, we introduce working distinctions between influence, status, prestige and esteem, linking influence and prestige to spectral ranking over normalised citation matrices. We present a recirculation algorithm for computing prestige indicators as the spectral rank over a multi-partite citation network projected to journals and institutions. Using the journal and institution prestige indicators we explain how to compute authors' relative esteem. We demonstrate the approach using the field of Economics, Finance and Business, and discuss our findings about the use of this approach as a screen. We conclude with a discussion of the limitations and prospects of this family of methods.

2 Prestige, esteem and spectral ranking

Researcher publication rates (articles/year) vary by orders of magnitude due to many factors and are only weakly linked to the status of their oeuvres (Cole, S and Cole, J, 1967; Ioannidis et al, 2024). Publication citation rates (cites/article/year) also vary by orders of magnitude due to many factors (Tahamtan et al, 2016;

Tahamtan and Bornmann, 2019), including an aspect of status called 'influence' or 'popularity' (Bollen et al, 2006; Garfield, 1979). Similar observations can be made about the citation rates of journals and institutions accumulated through their constituent publications.

That *status* is more than popularity has long been recognised in sociology (e.g. Katz, 1953; Podolny, 2005). It will be useful in this work to have terms for two kinds of status indicator: *prestige* for journals and institutions, and *esteem* for researchers. (Davis and Moore, 1945; Goode, 2022, This distinction reflects discussions in, e.g., []). *Prestige* is a long-term, relatively stable status indicator that is largest for journals like *Nature* (Baldwin, 2019) and institutions like those at the top of global university rankings (Aguillo et al, 2010). *Esteem* is a short-term, fluid status indicator that could be perceived to rise if a senior researcher becomes tenured at a high-prestige university, and decline if they are censured for questionable research practices (Aronson and Linder, 1965; Hassanibesheli et al, 2017).

According to our usage, the long-standing ISI journal impact factor (JIF) introduced in Garfield (1955, 2006) is an indicator of *journal prestige*, especially within a discipline. JIF reports the average number of citations per article in a journal, counting citations with equal value and thus ignoring the nature of citing works or journals. Garfield (2006) explains that the JIF helps identify the core journals in a field, some by larger size and others by higher citation frequency (higher impact), and suggests that some authors use the JIF to help select an outlet their work since 'journals with high impact factors include the most prestigious' (p. 92). Note that Bollen et al (2006) argue that the JIF is an indicator of popularity not prestige, since it takes no account of the standing of referencing authors.

Many researchers have advocated alternatives to the JIF that take account of the context of a reference. Waltman and Van Eck (2010) categorise alternatives as (a) normalising the cited-side, (b) normalising the citer-side, (c) using an indicator other than citations/article, or (d) recursively weighting citations. For example, the citer-side adjusted Audience Factor (Zitt and Small, 2008) adjusts the JIF for inter-journal differences by boosting or diminishing the value of a journal's references in proportion to the average number of references per article before summing the weighted citations. This compensates for disciplinary differences in the length of reference lists and in the delay before work is cited, both factors that influence the JIF. Habibzadeh and Yadollahie (2008) propose citer/citing-side scaling that weights a reference by the quotient of the JIFs of citing and cited journal, rescaling this weight to the range 0.1 to 10 over a logit function of the JIF ratio.

The JIF and many variants focus initially on individual journals characterised by the number, references and citations of their constituent papers. Comparisons to establish the relative standing of journals are made after the JIFs are computed. An alternative view is that journals inhabit a network or social system where relative standing is determined recursively by the whole community. An early study based on this view (Narin, 1976; Pinski and Narin, 1976a; Narin et al, 1976) [hereinafter PN] measures the influence of a publishing unit (journal, department, institution, nation) as the stable distribution of 'influence weight' (a kind of prestige) that is exchanged through reference lists. Vigna (2016b) adopts the generic term *spectral ranking* in a helpful summary of the antecedents and followers of PN.

PN provide a general approach to using citation matrices to spectrally rank *units* such as authors, journals, institutions, or countries. Let $c_{i,j}$ be the number of references given from unit i in the census time interval T_c and received by the unit j in the target time interval T_t. Since units publish different number of articles per year, and articles have reference list of different lengths, PN normalise the exchange from unit i to j as the ratio $c_{i,j}/s_j$ where the denominator $s_j = \sum_k c_{k,j}$ is the count of all references given out by j in T_c. The final *influence per reference* of each unit, w_i, is then determined recursively as the influence-weighted sum of the normalised ratios, $w_i = \sum_k w_k c_{i,k}/s_k$, with the scaling of w set by the condition that the average size-weighted influence is unity, $\sum_i s_i w_i = \sum_i s_i$. The *influence per article* is $v_i = \sum_k v_k (c_{i,k}/s_k)(a_k/a_i)$ where a_i is the number of articles published in T_c by journal i (see also Palacios-Huerta and Volij, 2004). The Index of New Knowledge (INK) proposed by Pauly and Stergiou (2008) is the initial, non-recursed stage of the Pinski-Narin algorithm.

Palacios-Huerta and Volij (2004) [PHV] adopt an axiomatic approach to the unit ranking problem (their work considers only journals as the unit, but this is not required). Defining reference intensity as the mean number of references per article in a unit, PHV show that the PN algorithm for the influence *per publication* (v_i above) uniquely satisfies four axioms: (1) the influence ratio is not affected by the reference intensity, (2) for unit of equal size and reference intensity the influence ratio is equal to $C_{i,j}/C_{j,i}$, (3) in a set of units having the same reference intensity, the removal of a unit does not change the influence ratio, and (4) splitting a unit into subsets does not change the influence ratio. Serrano (2004) offers an impossibly result to counter PVH's optimism, showing that the inclusion of a large unit with low citation flow to and from the original unit set markedly shifts v_i. Waltman and Van Eck (2010) draw the same conclusion by noting that the PN method is not invariant to disciplinary differences, a point also emphasised by PN.

PN used the power iteration to determine the principal eigenvector of the input-output matrix, and applied the method to rank separately Physics and Biology Journals. Chapter 10 of Narin (1976) also presents an application of the method to institutional influence ranking, but does not implement spectral ranking of institutions as units. Rather, the procedure first constructs journal sets and their influence weights (11 fields) and then derives the total influence of around 135 institutions (broadly and by field) by summing the institutional publication count in each journal weighted by the respective influence weight per publication. The algorithmic institutional influences compare favourably with rankings derived from surveys.

Motivated by this work, Geller (1978) considered an alternative exchange of influence or prestige through the ratio of the number of references from i to j over the total number of references from i, $\sum_k C_{ik}$. Geller's normalisation yields a row-stochastic matrix, the transition matrix of a Markov chain. Its principal eigenvector is the long-run probability distribution of *influence per citation* to each unit. Bollen et al (2006) propose a weighted pageRank algorithm that starts with the same row-normalising citation matrix, and follows Page (1998) by introducing an attenuation factor $0 \leq \alpha \leq 1$ and a minimum amount of influence (prestige) for each unit. Clarivate's Eigenfactor journal score (Bergstrom, 2007; West et al, 2010) and the Scimgo-Scopus journal ranking (González-Pereira et al, 2010; Guerrero-Bote and Moya-Anegón, 2012) also use the row-normalised form, with variants that exclude journal self-citations and/or adopt different time windows. Vigna (2016a) and Waltman and Van Eck (2010) survey connections between historical antecedents of recursive algorithms, and reveal the important connection between recursions and path summations.

2.1 Prestige for journals and institutions (JIRC algorithm)

The PN influence per reference w_i is a measure of the influence a citation from i relative to the average, and the ratio w_i/w_j is a measure of the relative influence of a citation from unit i and from unit j. Differences between unit size and reference density are removed. Accordingly, we will herein call w_i the *prestige* of unit i and use its value to represent the *esteem* that a work acquires from the citation.

We now extend PN to a two-unit model, specifically sources and institutions. The entities required to build this model comprise sets of N works, M sources, P authors and Q institutions. Works have a reference list to earlier <u>referenced</u> works, corresponding to a list of tuples $(w_i, w_j, 1)$ that specify the edge list of a binary directed work-work network $(\mathcal{W} \times \mathcal{W})$ with adjacency matrix (note that C is no longer the unit citation matrix)

$$\mathbf{C}^{\mathcal{W}\times\mathcal{W}} \in \{0,1\}^{N\times N} \quad c_{i,j} = \mathbf{1}[\text{work } i \text{ references work } j] \tag{1}$$

The scholarly communication system forms a multipartite network where $\mathbf{C}^{\mathcal{W}\times\mathcal{W}}$ links other entity types through bipartite projections:

$$\mathbf{C}^{\mathcal{S}\times\mathcal{I}} = \mathbf{B}^\top \mathbf{C}^{\mathcal{W}\times\mathcal{W}}(\mathbf{FG}) \tag{2}$$

$$\mathbf{C}^{\mathcal{I}\times\mathcal{S}} = (\mathbf{FG})^\top \mathbf{C}^{\mathcal{W}\times\mathcal{W}}\mathbf{B} \tag{3}$$

where

$$\mathbf{B} \in \{0,1\}^{N\times M}, \quad B_{ws} = \mathbf{1}[\text{work } w \text{ published in source } s] \tag{4}$$

and either (institution fractionation)

$$\mathbf{F} \in \mathbb{R}^{N\times N}, \quad F_{ww} = \text{diag}(1) \tag{5}$$

$$\mathbf{G} \in \mathbb{R}^{N\times Q}, \quad G_{wi} = \frac{\mathbf{1}[\text{institution } i \text{ is affiliated with work } w]}{n_i(w)} \tag{6}$$

or (author fractionation)

$$\mathbf{F} \in \mathbb{R}^{N\times P}, \quad F_{wa} = \frac{\mathbf{1}[\text{author } a \text{ contributed to work } w]}{n_a(w)} \tag{7}$$

$$\mathbf{G} \in \mathbb{R}^{P\times Q}, \quad G_{ai} = \frac{\mathbf{1}[\text{author } a \text{ affiliated with institution } i]}{n_i(a)} \tag{8}$$

Here, $n_a(w)$, $n_i(w)$, and $n_i(a)$ for work i are respectively the number of authors, institutions, and institutions per author. The full journal-institution citation has the block form

$$\mathbf{K} = \begin{pmatrix} \mathbf{B}^\top\mathbf{C}\mathbf{B} & \mathbf{B}^\top\mathbf{C}(\mathbf{F}\mathbf{G}) \\ (\mathbf{F}\mathbf{G})^\top\mathbf{C}\mathbf{B} & (\mathbf{F}\mathbf{G})^\top\mathbf{C}()FG \end{pmatrix} \to \begin{pmatrix} \mathbf{0} & \mathbf{C}^{\mathcal{S}\times\mathcal{I}} \\ \mathbf{C}^{\mathcal{I}\times\mathcal{S}} & \mathbf{0} \end{pmatrix} \tag{9}$$

However, we omit source-source ($\mathbf{B}^\top\mathbf{C}\mathbf{B}$) and institution-institution ($(\mathbf{F}\mathbf{G})^\top\mathbf{C}()FG$) links, in the expectation that this will minimise any adverse effects of source and institution 'self-citations.'

3 Journal and Institution Influence

The article sets $\mathcal{P}^{\mathcal{C}}$ and $\mathcal{P}^{\mathcal{T}}$ associated with the journal set $\mathcal{J}$ involve authors and institutions listed in the article affiliation metadata.

The dataset now contains the ensemble of journals, $j = 1 \ldots M$, as well as a collection of institutions, $i = 1 \ldots N$ with a sizable number of papers in the field and period under analysis.

When computing journal influence scores, JIRC will inject values of citing article prestige, v_i, based on institutions assessments derived from the dataset. Conversely, to computing institutional influence scores, JIRC will inject values of citing article prestige, w_j, based on journal assessments derived from the same dataset. We will formulate the problem as a (re)circulation of reputation. The idea is to go from institutions to institutions through journals and vice versa.

The problem can be stated as follows.

Let's introduce citation matrix A, from institutions ($i = 1 \ldots N$) to journals ($j = 1 \ldots M$), $A = (c_{ij})$, whose elements are the sum of references emitted by the articles published by institution i (rows) that become citations received by any of the articles published in journal j (columns) during the target publication window.

Let's again define the overall sum of all references from institution i as

$$s_i = \sum_{j=1}^{M} c_{ij} \tag{10}$$

The matrix A can be normalised as the row stochastic matrix P_A, where

$$P_A = (a_{ij}) = (c_{ij}/s_i) \tag{11}$$

in which case references from an institution are normalized by the total number of references from that institution to all journals in the dataset.

We also introduce citation matrix B, from journals to institutions, $B = (c_{ji})$, whose elements are the sum of references emitted by the articles published by journal j (rows) that become citations received by any of the articles published in institution i (columns) during the target publication window.

$$s_j = \sum_{i=1}^{N} c_{ji} \tag{12}$$

The matrix B can be normalised as the row stochastic matrix P_B, where

$$P_B = (b_{ji}) = (c_{ji}/s_j) \tag{13}$$

in which case, references from a journal are normalized by the total number of references from that journal to all the institutions in the dataset.

The influence vectors are connected by the two matrices. P_A^T is $M \times N$, whereas P_B^T is $N \times M$. Both influence column vectors, w, $M \times 1$, for journals and v, $N \times 1$, for institutions, receive reputation shots from each other, $w = P_A^T v$, $v = P_B^T w$.

Hence, we can now treat the reputation transfer full circle, as

$$v = P_B^T P_A^T v \tag{14}$$
$$w = P_A^T P_B^T w \tag{15}$$

Matrices $P_A^T P_B^T$ and $P_B^T P_A^T$ are both square, although they have, in general, different sizes: $M \times M$ and $N \times N$, respectively.

However, it is well-known that they share all the nonzero eigenvalues, the largest in particular, implying that once we find the eigenvector v corresponding to the largest eigenvalue of $P_B^T P_A^T$, we know that the eigenvector corresponding to the largest eigenvalue of $P_A^T P_B^T$ will be $w = P_A^T v$, and vice versa.

Equations $v = P_B^T P_A^T v$ and $w = P_A^T P_B^T w$ define the influence weight for institutions and journals, respectively. The influence weights per publication, v' and w' are:

$$v'_i = \frac{(P_B^T P_A^T v)_i}{b_i}$$
$$w'_j = \frac{(P_B^T P_A^T w)_j}{a_j}$$

where b_i and a_j are the number of articles published in the selected field and time period by institutions and journals, respectively.

JIRC will also use the Power method either for institutions or journals, since both will produce the same result:

$$v(k) = \frac{P_B^T P_A^T v^{(k-1)}}{|P_B^T P_A^T v^{(k-1)}|} \tag{16}$$

that converges to the eigenvector, v, associated with the largest eigenvalue, and the desired normalized influence vector, v'. The equation $w = P_A^T v$ will render the normalized eigenvector for the journal to journal transfer of reputation,

$$w'_j = \frac{(P_A^T v)_j}{a_j} \tag{17}$$

4 Methods

4.1 Data Collection and Preparation

Our analysis is based on publication data from the Web of Science (WoS) Core Collection, with sizable number of documents (article and review type) classified within the ESI Field Economics & Business in the year 2020. This field encompasses four WoS categories: Business, Economics, Finance, and Management.

The dataset includes:

- Journals (J): $M = 613$ journals focused on Economics & Business.
- Institutions (I): $N = 600$ institutions worldwide with a substantial number of publications in the ESI field Economics & Business.
- Published Articles (PA): More than $41,000$ articles published in 2020, classified within the Economics & Business ESI Field by the Clarivate motor InCites, including journal names and institutional affiliations.
- Citing Articles (CA): More than $200,000$ articles published between 2020 and 2025 classified within the Economics & Business ESI Field by the Clarivate motor InCites, citing any of the articles belonging to PA, including their full citation data and institutional affiliations.

4.2 Matrix Construction: The JIRC Framework

As explained before, the core of our method involves the construction of two citation matrices that form a closed loop for reputation recirculation.

- Matrix A (Institutions → Journals): an (N × M) matrix where each element a_{ij} is the total number of citations from all articles in CA published by institution i to all articles in PA published in journal j.
- Matrix B (Journals → Institutions): an (M × N) matrix where each element b_{ji} is the total number of citations from all articles in CA published in journal j to all articles published in PA by institution i.

Following the Pinski-Narin methodology, we normalize these matrices to be row-stochastic, creating transition matrices P_A and P_B (as defined in Eqs. 11 and 13 in the introduction). This normalization controls for the size of institutions and journals, ensuring we measure average influence rather than raw volume.

4.3 Solving the Reputation Recirculation System

The influence vectors for institutions, v, and journals, w, are defined by the coupled system:

$$v = P_B^T P_A^T v$$

$$w = P_A^T P_B^T w$$

To check the irreducibility of the two square matrices $P_B^T P_A^T$ and $P_A^T P_B^T$, we rely on a standard result in the field of matrix theory, connected to Perron-Frobenius theory, which states that a square matrix is irreducible if and only if its associated directed graph is strongly connected. We used an algorithm based on Dijkstra (1959) to settle the issue. Afterwards, we solve for the principal eigenvector v of the (N × N) matrix $P_B^T P_A^T$ using the power method (Eq. 16). The journal influence vector is then directly calculated as $w = P_A^T v$.

4.4 Researcher Screening Application

To validate the practical utility of JIRC in individual research evaluation, we assembled a list of 330 prominent researchers in Economics & Business. This list was split into two groups:

1. Control Group (C): 165 Researchers with universally acknowledged influence, including recipients of the Nobel Prize in Economics, the John Bates Clark Medal, Gossen Prize, and other prestigious field-specific awards, as well as European Research Council advanced grant holders.
2. Test Group (T):165 Researchers with a high number of highly-cited papers but without the prestigious awards defining the Control group.

For each paper published by any of the researchers from our sample, we computed a JIRC score as the product of the journals's JIRC score and the best institutional JIRC score among all the institution in the paper byline

Then, for each researcher, we calculated a personal JIRC score by averaging the JIRC scores of all their publications between 2015 and 2024 (matching the temporal window used by Clarivate in 2025 to identify highly cited papers). Researchers were then ranked and divided into three tiers based on this averaged score.

5 Results

5.1 Journal and Institution Rankings

The JIRC algorithm produced stable and convergent influence scores for all journals and institutions. The journal rankings ([Table/Appendix 1]) showed strong concordance with expert judgment and other reputation-based rankings, while also introducing nuance by factoring in the prestige of citing institutions. Similarly, the institutional rankings ([Table/Appendix 2]) reflected global academic reputations, effectively identifying centers of genuine intellectual influence.

5.2 Disentangling Researcher's Influence from Notoriety

The results of the researcher screening through individual JIRC scores were striking and demonstrate the core strength of the method.

- Tier 1 (Top 100 Researchers): This group was mainly composed of researchers from the Control group (88%). These are individuals whose work is not only widely cited but is cited by influential entities (journals and institutions).
- Tier 2 (Middle 130 Researchers): This group showed a mix of both Control and Test group members (75 vs 55), indicating a spectrum of influence among highly cited authors.
- Tier 3 (Bottom 100 Researchers): This group was overwhelmingly populated by researchers from the Test group (98%). This suggests that while these individuals have achieved high citation counts (notoriety or popularity), their work is cited by sources with lower aggregate influence scores, potentially indicating strategic or circular citation practices within specific sub-communities, or work that is popular but not foundational.

6 Discussion

Our proposed JIRC algorithm successfully extends the eigenvector centrality paradigm beyond the journal-journal citation network. By recirculating reputation through a journal-institution loop, we ground journal influence in the prestige of its citing institutions and, conversely, ground institutional prestige in the influence of its citing journals.

The most significant finding is the algorithm's ability to differentiate between influence and mere popularity at the individual researcher level. The clear separation between the award-winning Control group and the highly-cited-but-unawarded Test group across the tiers provides strong empirical validation. It suggests that JIRC is resistant to inflation from tactics like citation stacking or publishing in journals that prioritize rapid publication and high citation volume over intellectual rigor and influence.

The algorithm's accuracy is dependent on clean, disambiguated data for institutions. Future work will explore the impact of self-citations at the institutional and journal level, and the application of JIRC to other scientific fields and to author-level networks directly.

7 Conclusion

The Journal-Institution Reputation Circulation algorithm offers a robust and theoretically grounded framework for assessing scholarly influence. It moves beyond counting citations to weighing them based on a recursive definition of reputation that flows between producers (institutions) and disseminators (journals) of knowledge. Our results demonstrate that JIRC is not just a metric for ranking journals and institutions but also a powerful tool for research evaluation, capable of screening individual researchers to identify those with genuine, influential impact as opposed to those who have gained notoriety from a purely citation-based system. In an era of increasing publication volume and potential metric manipulation, JIRC provides a nuanced and reliable measure of true scholarly prestige.

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